

1. Administrative & Study Identification

Full Study Title: A Phase II, randomized, open-label, multi-center study of JSB462 (luxdegalutamide) in combination with abiraterone in adult male patients with metastatic hormone-sensitive prostate cancer (mHSPC) **Short Title/Acronym:** LuxAR-03 **Protocol Number:** CJSB462C12201 **NCT Number:** NCT06991556 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** JSB462 (Luxdegalutamide) + Abiraterone **Phase of Development:** Phase II **Trial Status:** RECRUITING **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To select the recommended dose of the combination of luxdegalutamide and abiraterone for phase III by evaluating the efficacy, safety, and tolerability in patients with mHSPC. Towards that end, the totality of the efficacy, safety, tolerability and PK data from participants randomized in the study will be evaluated. **Secondary Objectives:** To evaluate radiographic progression-free survival (rPFS), overall survival (OS), PSA0 (PSA level <0.2 ng/mL), pharmacokinetics (PK), and safety (incidence rate of adverse events) across all study treatments.

2.2 Background & Rationale To address the rapid progression to metastatic castration-resistant prostate cancer (mCRPC) in patients with high-volume mHSPC. While androgen deprivation therapy (ADT) combined with an androgen receptor pathway inhibitor (ARPI) is the standard of care, patients with high-volume disease have poorer outcomes. Luxdegalutamide is a novel oral proteolysis-targeting chimera (PROTAC) androgen receptor (AR) degrader. Preclinical and Phase I data demonstrate that targeting the AR at multiple steps by combining luxdegalutamide with abiraterone provides complementary antitumor activity, aiming to induce deeper PSA responses and delay disease progression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Abiraterone or Enzalutamide) **Blinding:** Open-label **Randomization:** 1:1:1 Stratified

3.2 Planned Enrollment & Duration Target \$\$\$: 150 patients **Number of Sites:** Multi-center (Global) **Estimated Study Start:** 2024/2025

Estimated Primary Completion: Upon achievement of required progression events and PSA tracking endpoints.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult male patients with histologically confirmed adenocarcinoma of the prostate. 2. High-volume mHSPC, defined by the presence of ≥ 1 metastatic visceral non-nodal lesion and/or ≥ 4 metastatic bone lesions (with at least one lesion outside the vertebral column and/or pelvis) in imaging exams obtained ≤ 28 days prior to randomization. 3. Eastern Cooperative Oncology Group (ECOG) performance status ≤ 2 . 4. Castrate level of serum/plasma testosterone (<50 ng/dL or <1.7 nmol/L). 5. Ongoing ADT (prior orchiectomy and/or ongoing GnRH analog/antagonist) for ≤ 90 days prior to randomization is allowed, provided that PSA zero (PSA level <0.2 ng/ml) is not achieved prior to randomization.

4.2 Exclusion Criteria 1. Prior exposure to a second-generation ARPI (enzalutamide, darolutamide, apalutamide, and/or abiraterone) for the treatment of advanced/metastatic disease. (Note: Prior exposure to ARPI, taxane chemotherapy up to 6 cycles, or radioligand therapy in the neo/adjuvant localized setting is allowed if the last dose was >12 months from randomization. Prior use of 1st-generation ARPI like bicalutamide for ADT initiation is allowed if ≤ 14 days and last dose was ≥ 7 days from randomization). 2. Participants with mixed histology (neuroendocrine) prostate cancer. 3. Participants with biochemical recurrence only or those without evidence of metastatic disease by radiological imaging (CT/MRI or bone scan).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Arm 1:

Luxdegalutamide 100 mg once a day (QD) + Abiraterone 1000 mg QD.

* **Arm 2:** Luxdegalutamide 300 mg QD + Abiraterone 1000 mg QD. *

Arm 3 (Control): Abiraterone 1000 mg QD OR Enzalutamide 160 mg QD (Investigator's choice).

Route of Administration: Oral **Duration of Treatment:** Administered continuously until disease progression (per PCWG3-modified RECIST 1.1), unacceptable toxicity, death, investigator decision, or participant decision.

5.2 Concomitant & Prohibited Therapy Permitted: Concurrent continuous Androgen Deprivation Therapy (ADT). **Prohibited:** Concurrent investigational therapies or additional targeted oncology treatments outside the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Baseline Imaging (CT/MRI/Bone Scan), PSA measurement, Histology review.
Baseline	Day 0	Randomization (1:1:1), First Dose.
Interim / Treatment Period	Continuous	PSA testing, CTCAE/PRO-CTCAE Symptom Questionnaires, Safety Labs, Radiographic Response tracking until progression/toxicity.
Safety Follow-up	30 Days (+/- 7 days) post-treatment	Treatment Discontinuation, 30-day Safety Follow-Up visit.
Long-Term Follow-up	Post Safety Follow-up until Study End	Collection of safety, efficacy, and survival information.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Prostate Specific Antigen 90 (PSA90)

Rate: Defined as the proportion of participants achieving a $\geq 90\%$ decrease in PSA from baseline at any timepoint, confirmed by a second PSA measurement ≥ 3 weeks later without any PSA progression in between. 2. **Incidence rate of adverse events (AEs):** Categorized by type, frequency, and severity, as graded by the NCI CTCAE version 5.0. 3. **Number of participants with dose adjustments:** Dose reductions, interruption, or permanent discontinuation, summarized by treatment arm.

7.2 Secondary & Exploratory Endpoints * Radiographic Progression-Free Survival (rPFS). * Overall Survival (OS). * Incidence rate of adverse events (AEs). * PSA0 Rate (PSA level < 0.2 ng/mL at any timepoint). * Pharmacokinetics (PK).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Evaluated through standard NCI CTCAE version 5.0 clinical grading and the PRO-CTCAE to capture symptom frequency, severity, and interference. **Serious Adverse Events (SAEs):** Reported promptly following global safety guidelines. **Data Monitoring Committee (DMC):** Utilized for ongoing safety and tolerability assessment across the treatment arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy. **Sample Size Justification:** $N=150$ randomized 1:1:1. Powered using the stratified Miettinen and Nurminen method to obtain the minimum response rates required for the lower bound of the 90% confidence interval of the PSA90 difference to exceed 0% (clinical benefit) and 10% (strong clinical benefit) between the pooled experimental arms and the control arm. **Handling of Missing**

Data: Standard oncology trial multiple imputation frameworks for missing continuous variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine clinical monitoring for safety signals, dose adjustments (reductions, interruptions, or permanent discontinuations), and treatment discontinuations. **Audit Trail:** Centralized independent review of radiographic imaging (PCWG3-modified RECIST 1.1) and verifiable eCRF systems.

11. Study Identifiers & Governance

Primary ID: CJSB462C12201 **Registry Number:** NCT06991556 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** LuxAR-03 **Official Regulatory Title:** A Phase II, Randomized, Open-label, Multi-center Study of JSB462 (Luxdegalutamide) in Combination With Abiraterone in Adult Male Patients With Metastatic Hormone-sensitive Prostate Cancer (mHSPC)

12. Clinical Framework (mHSPC Specific)

Primary Condition: Metastatic Hormone-Sensitive Prostate Cancer (mHSPC) **Diagnosis Threshold:** High-volume disease (≥ 1 metastatic visceral non-nodal lesion and/or ≥ 4 bone lesions with at least one lesion outside the vertebral column and/or pelvis) **Medical Methodology:** Androgen Deprivation Therapy (ADT) combined with an ARPI and PROTAC AR degrader **Optimization Target:** Deepest possible PSA suppression (PSA90 and PSA0) and delayed radiographic progression

13. Study Procedures & Methodology

Management Pathway: Standard oncology treatment pathway modified for combination oral targeted therapy **Education Protocol (HCP):** Site initiation visits and regular protocol safety monitoring updates **Education Protocol (Patient):** PRO-CTCAE patient-reported outcomes measurement system training for self-reporting of symptoms **Quality Audit Mechanism:** Independent central imaging review and stringent tracking of treatment adherence

14. Observation & Follow-Up Schedule

Total Duration: Continuous dosing until disease progression, toxicity, or participant/investigator decision; followed by a 30-

day (+/- 7 days) safety follow-up, and subsequent long-term follow-up until the end of the study. **Onsite Visit Cadence:** Pre-defined oncological cycle assessments (e.g., Q4W/Q8W depending on cycle length) **Remote Monitoring Frequency:** Routine telehealth check-ins and PRO-CTCAE surveys **Checklist Review Interval:** Prior to the initiation of each treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Dr. Michael Schweizer / Dr. Rana McKay]	_____	[DD-MMM-YYYY]	Clinical Operations Lead			
[Novartis Rep. Name]	_____		[DD-MMM-YYYY]	Lead Statistician			
[Name]	_____		[DD-MMM-YYYY]				